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REVIEW

Multidrug-resistant *Acinetobacter baumannii* infections: looming threat in the Indian clinical setting

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ABSTRACT

Introduction: The recent increase in multidrug-resistant strains of *A. baumannii* has increased the incidences of ventilator-associated pneumoniae, catheter-associated urinary tract infections, and central line-associated blood stream infections, together increasing hospital stay, treatment cost, and mortality. Resistance genes *blaOXA and blaNDM* are dominant in India. Carbapenem-resistant *A. baumannii* (*CRAB*) International clone-2 (IC-2) are rising in India. High dependency on carbapenems and last-resort combination of tigecycline and polymyxins have aggravated outcomes. Despite nursing barriers, ward closure, environmental disinfections etc for detecting and controlling transmission, MDR isolates and *CRAB* nosocomial outbreaks continue. Treatment cost overruns by AMR adversely affect 80% of Indians without insurance cover.

Area covered: This narrative review will cover epidemiology, resistance pattern, genetic diversity, device-related infection, cost, and mortality due to multidrug-resistant and *CRAB* in India. A comprehensive literature search in PubMed and Google Scholar using appropriate keywords at different time points yielded relevant articles.

Expert opinion: It is challenging to enforce policies to control MDR *A. baumannii* in India. Government and hospitals should enforce stringent infection control measures, surveillance, and antimicrobial stewardship to prevent further spread and emergence of more virulent and resistant strains. Knowledge on antibiotic resistance mechanisms can help design novel antibiotics that can evade, resistance.

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1. Introduction

Acinetobacter baumannii complex (*A. b* complex) is a group of pathogens that includes *A baumannii, A. calcoaceticus, A. nosocomialis,* and *A. pittii* [1]. Of the 60 known *Acinetobacter* spp.,*A. b* complex is of great clinical significance, and is associated with a wide spectrum of hospital-acquired infections (HAI) mostly prevalent among patients in intensive care units (ICU). *A. b* complex cause a wide range of infections, such as community and hospital-acquired pneumonia, skin and soft tissue infections, bloodstream infections (BSI) and urinary tract infections (UTI) [2].

Within *A. b* complex, *A. baumannii* is the most common and mounting cause of nosocomial infection and has become a significant hospital pathogen, due to emergence of multi-drug-resistant (MDR) strains. Today, *A. baumannii* is considered one of the six most essential micro-organisms that cause HAI world-wide, with attributable mortality ranging from 8% to 35%, depending on strain and type, increasing hospital stay and health care expenditure [3,4].

Due to the ability to survive in a harsh environment (wide range of temperatures and pH, presence of disinfectants, exposure to antimicrobials), *A.b complex* infections persist and can spread to different healthcare facilities by colonizing or infecting patients [5]. Pathogens under *A. b* complex possess both acquired and intrinsic resistance to many common antimicrobials, such as aminoglycosides (AGs), cephalosporins and penicillins. Consequently, last-resort antimicrobials such as carbapenems have become the key treatment option for *A. b* complex infections [6]. However, resistance against carbapenem has been increasingly reported world-wide, ever since the discovery of imipenem in 1985 (first reported in 1985 in patient at Royal Infirmary of Edinburgh). Infections by *carbapenem-resistant A. baumannii* (*CRAB*) have emerged as an unprecedented therapeutic challenge with a high degree of morbidity and mortality. MDR *A. baumannii* is a major cause of outbreaks in hospitals and associated with widespread clonal dissemination. The spread of such highly virulent clones has increased MDR *A. baumannii* infections globally [7].

The world witnesses an annual incidence of 100,000 *A. baumannii* cases, out of which 50% show resistance to multiple antimicrobials, including carbapenem [8]. Both World Health Organization (WHO) and the Centers for Disease Control and Prevention (CDC) have listed *CRAB* as an AMR pathogen of high priority. In 2017, the USA reported over 8,500 inpatient *CRAB* infections leading to

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Article highlights

- Multidrug-resistant *A. baumannii* infections are increasing across Indian healthcare settings.
- Carbapenem-resistant *A. baumannii* (*CRAB*) are of grave significance to public health world-wide because of their association with high treatment costs, morbidity, and mortality.
- CRAB strains show resistance toward last-resort antibiotics such as tigecyclines and polymyxins.
- Studies from India have shown considerable diversity in antimicrobial resistance genes related to β-lactamases, expression of efflux proteins, persistence on surfaces, and extensive biofilm production in multiple strains.
- The infection control measures in India are sluggish, thereby increasing device-related infections in ICU's and other hospital settings.
- Antimicrobial stewardship programs must be intensified to avoid escalation of highly resistant *A. baumannii* infections, potentially taking us closer to the post-antibiotic era.

700 deaths. More than 50% of *A. baumannii* infections traced to ICUs in the USA and Europe are *CRAB* [8,9]. Indian ICUs especially from Chandigarh and Hyderabad have also reported 12–41% MDR cases [10]. Alarmingly, India has witnessed an overall *CRAB* cases of 40% to 85% that includes wards and ICUs of the hospitals from the states of Tamilandu, New Delhi and Punjab [11,12]. International Nosocomial Infection Control Consortium (INICC) surveillance study between 2004–2009 for 36 countries, including India, summarized a high proportion of device-associated-hospital-acquired infections (DA-HAI), including ventilator-associated pneumonia (VAP), central line-associated bloodstream infection (CLABSI) and catheter-associated urinary tract infection (CAUTI) due to *CRAB* [13]. *CRAB* is a major opportunistic pathogen in hospital settings. To our knowledge, there are no significant reports of *CRAB* in the community in India [14].

MDR and *CRAB* infections persist and can spread to different healthcare facilities by colonizing or infecting patients [5,15]. Many case–control studies suggest that prior antimicrobial exposure is a significant risk factor for colonization and infection. Carbapenems and third-generation cephalosporins (GCs) are most commonly associated with AMR, followed by fluoroquinolones (FQs), AGs, and metronidazole. Mechanical ventilation (MV) ranks second. The severity of illness, longer ICU and/or hospital stay, recent surgery, and invasive procedures come the next. Environmental contamination is also an important risk factor for MDR *A. baumannii* outbreaks [5]. Prevention of MDR and *CRAB* infections is a challenge [15]. While being an important nosocomial pathogen, community-acquired infections by this pathogen (including bacteremia and pneumonia) are less common but are associated with relatively high mortality. Most community-acquired infection cases were reported in Asian countries, especially from India. Risk factors for community-acquired infections caused by MDR strains of *A. baumannii* include diabetes mellitus, renal disease, chronic obstructive pulmonary disease, smoking, and excessive alcohol consumption [16].

2. Resistance pattern

A. b complex, being present in the soil, can evolve AMR rapidly because of long-term exposure to antibiotics produced by soil microbes. Other micro-organisms take longer to acquire resistance in response to antimicrobial therapy. AMR has emerged in *Acinetobacter* spp. via selection pressure exerted by broad-spectrum antimicrobials and transmission of resistant strains between patients [5].

A. baumannii is naturally resistant to multiple antibiotics because of simultaneous and concerted processes such as:

(a) Low permeability of porins in the outer membrane and its small size. The presence of Outer membrane protein-A (OmpA) makes the porins smaller [17].

(b) Overexpression of efflux pumps such as AbeABC, AbeM, AbeIJK, AbeFGH, Tet(A), and Tet(B), causing widespread resistance to cephalosporins, FQs, tetracyclines, AGs, and tigecycline.

(c) Inherent expression of β-lactamases such as AmpC, cephalosporinases, OXA-like β-lactamases, and carbapenemases

(d) Presence of a 'resistance Island' encompassing multiple resistance genes associated with many mobile elements.

(e) Potential for the horizontal acquisition of foreign resistance elements, for example, OXA-23, NDM (New Delhi Metallo-β-lactamase) carbapenemases, and aminoglycoside-modifying enzyme.

(f) Loss of lipopolysaccharide production due to mutations in lipid A biosynthesis genes such as *lpxA*, *lpxC*, and *lpxD* and overexpression of ISAba1-driven eptA has led to colistin-resistant strains [18].

(g) Production of biofilms, facilitated by an exopolysaccharide known as K-locus capsular polysaccharide and acyl-homoserine lactone-signaling molecules, which allows well-coordinated cell–cell interaction ('quorum sensing') [6,8,19,20].

A. baumannii also expresses a powerful cytotoxin, targeting the host's epithelial cells, thereby promoting *A. baumannii* colonization. *CRAB* infections feature the production of carbapenem hydrolyzing enzymes such as AmpC, OXA-like β-lactamases, and carbapenemases. *CRAB* has emerged rapidly with a 30% increase between 1995 and 2004. This coincides with the rapid spread of β-lactamases enzyme oxacillinase (OXA). Meanwhile, *A. baumannii* also developed resistance rapidly to AGs, FQs, and piperacillin-tazobactam. MDR *A. baumannii* outbreaks are extremely dangerous and have forced ward closures in many hospitals in the USA and Europe [6].

Since Czech Republic first reported colistin-resistant *A. baumannii* in 1999, a global spread has been evoking concern. The United Kingdom [2%] and the USA (5.4%) have reported low resistance levels, whereas other countries such as Korea (30%), Spain (40.7%), Iran (48%), and India (53%) have reported higher levels of colistin resistance [21].

2.1. *Prevalence of MDR* A. baumannii *and their resistance pattern in Indian health care settings*

Studies from 2005 and 2006 have reported 14% and 35% *CRAB* isolates from India, respectively [5]. In 2015, Gandra et al.

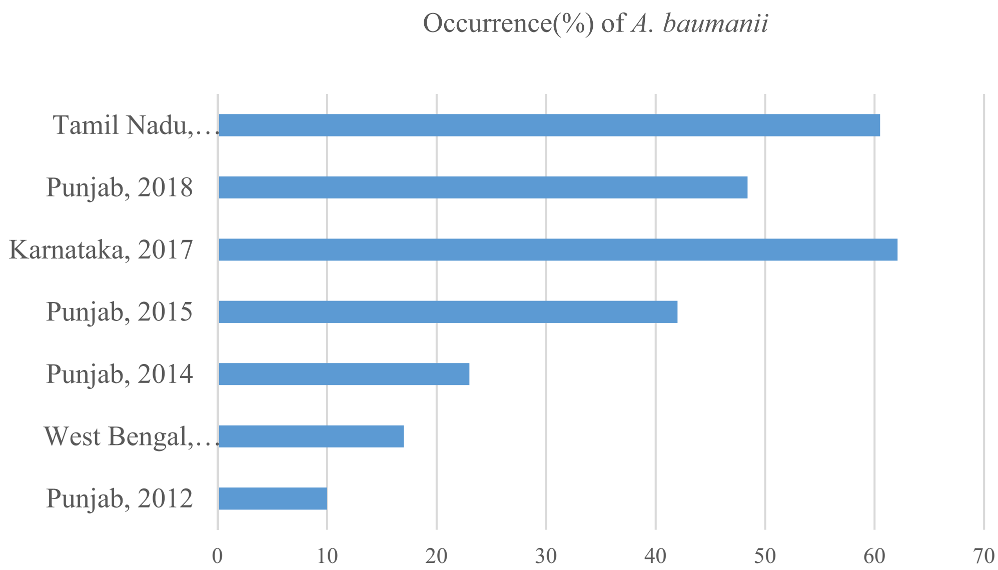

Figure 1. Rising occurrence of *A. baumanii* resistance across India: Timeline.

Table 1. Distribution of MDR, XDR, and PDR strains of *A. baumannii* across India.

Author and year	Location	MDR	XDR	PDR
Khamari et al. 2020[37]	Andhra Pradesh	21	64	-
Kalal et al. 2020[15]	Karnataka	60	40	-
Tewari et al. 2018[24]	New Delhi	87	-	-
Kaur et al. 2018[27]	Punjab	>95	>50	-
Vekataraman et al. 2018[42]	Tamil Nadu, Maharashtra, New Delhi, Haryana, Telangana	88	-	-
Banerjee et al. 2018[28]	West Bengal	88	62	-
Sudhaharan et al. 2015[39]	Telangana	77	13	-
Rynga et al. 2015[40]	Delhi	91	78	2
Gupta et al. 2015[41]	Pune	40	-	-
Bhattacharya et al. 2013[38]	West Bengal	44	-	-
Mathai et al. 2012[10]	Punjab	70	-	-

The numbers are expressed in % and are rounded to the nearest 10. MDR, multidrug-resistant; XDR, extensive drug resistant; PDR, pan-drug resistant.

reported an overall *CRAB* prevalence of 10% in 10 different hospitals across India [22]. Kumar et al. in 2019 have shown a prevalence of 89% for *CRAB* [23]. Many studies between 2012 and 2020 showed a varying prevalence MDR *A. baumannii* ranging from 0.4% to 62.1% (Figure 1)[23–37], suggesting the rising prevalence over the last eight years. Reports from different parts of India suggest 13 to 78% Increase in MDR *A baumannii* (>95%) and extensive drug resistant (XDR) in *A. baumannii* isolates (Table 1) [10,15,24,27,28,37–41]. A multi-center observational study by Indian Society of Critical Care Medicine (MOSER study) in 2018 found 88% of MDR *A. baumannii* [42]. The resistance pattern among *CRAB* toward various antimicrobials widely varied between different studies in India between 2009 and 2020 (Table 2). For instance, 93% *A. baumannii* stains showed resistance to penicillin-β-lactamase combinations such as amoxicillin-clavulanic acid and piperacillin-tazobactam [24,25]. Reports suggest that 85% and 97% of *A. baumannii* isolates are resistant to cephalosporins and β-lactamase combinations such as Cefoperazone-sulbactam [15,27]. Ninety-two percent of isolates containing *A. baumannii* were resistant to AGs such as amikacin and gentamicin [25]. A few studies between 2009 and 2020 in India showed that more than 90% of the isolates (*A. baumannii*) were *CRAB*, probably implying that carbapenems would no longer qualify as the last-resort antimicrobial (Figure 2) [15,25,43]. In *CRAB* isolates, tigecycline and polymyxins remain the last-resort prescriptions today. Furthermore, studies show an increasing trend in the resistance rates of these isolates toward tigecycline and polymyxins. MDR *A. baumannii* isolates resistant to tigecycline increased from 0% to 42% between 2012 and 2015 [24,25,28,39,40]. Many reports from India point to inappropriately high use of 'Watch group' antibiotics, driving AMR, and forcing a preference for 'Reserve group' such as polymyxins and tigecycline [43–45]. Colistin resistant strains across India also increased from 0% to 60% between 2012 and 2016 [24–27,29,37,40,46]. Mathai et al. (2012) showed that 13% of MDR strains of *A. baumannii* were resistant to polymyxin-B [10]. However, subsequent studies showed all *A. baumannii* strains were susceptible to polymyxin-B, probably indicating diminished use of polymyxin-B as last-resort antimicrobials across India. *A. baumannii* showing colistin resistance is increasing globally [27]. A meta-analysis showed that the prevalence of colistin-resistant *A. baumannii* in South-East Asia and Eastern Mediterranean countries is higher than in other regions of the world. A recent study from India showed 8% to 11% of the clinical isolates of *A. baumannii* were colistin-resistant [18]. Therefore, policy should mandate developing and implementing appropriate antibiotic usage guidelines in these countries [47].

Table 2. Geographical distribution of *A. baumannii* resistance pattern toward different classes of antibiotics across India.

Author and year	Location	ICUs/Wards	Penicillins+ β-lactamases		Cephalosporins with or without β-lactamases					FQs			AGs		Carbapenem			Tet	Polymixins	
			Amox + clav	Piper + taz	1st gen	2nd gen	3rd gen	4th gen	Cefo + sulb	Cipro	Levo	TMP-SMX	Amika	Genta	Imp	Mrp	Tige	Mino	Colistin	Poly-B
Mathai et al. (2012)[10]	Punjab	ICU	-	-	100	-	59–96	-	-	-	-	-	81	-	-	80	80	-	-	13
Bhattacharya et al. (2013)[38]	West Bengal	Wards	-	18	-	-	75–80	-	-	52	21	72	56	-	15	-	-	-	-	-
Datta et al. (2014)[35]	Punjab	ICU	-	50	-	-	90–96	-	-	70	-	-	-	90	57	-	-	-	-	-
Rynga et al. (2015)[40]	Delhi	ICU, wards	65	85	-	100	100	-	100	98	-		97	95	85	-	42	-	3	-
Gupta et al. (2015)[41]	Pune	ICU, wards, out-patient	-	-	-	-	43–46	44	-	23	-	-	42		22	-	-	-	-	-
Ghanshani et al. (2015)[62]	Rajasthan	ICU	20.9%	-	-	-	-	-	29	-	-	-	-	-	-	-	4	-	1	0
Saranathan et al. (2015)[59]	Pondicherry	ICU	62%	-	-	73	21–88	-	-	75	5	-	79	-	51	78	-	-	36	11
Gupta et al.[a] (2016)[33]	Karnataka	ICU	-	97		100	100	100	22	97	-	96	94	100	96	100	50	-	53	-
Kaur et al (2016)[34]	Punjab	ICU, ward	83	85	-	-	98	96	-	93	-	94	87	83	43	55	-	-	4	2
Mahanthesh et al.[a] (2017)[26]	Karnataka	NICU, PICU	85	85	97	-	78–89	-	96%	87	-	-	75	-	68	65	-	-	60	-
Mahanthesh et al. (2017)[32]	Karnataka	PICU	-	52	-	-	72–88	46	-	67	-	37	63	63	46	55	-	-	-	-
Khan et al. (2017)[36]	Delhi, West Bengal, Maharastra	ICU	100	100	-	-	100	-	100	100	-	100	100	-	100	100	0% 1	-	0	-
Kaur et al. (2018)[29]	Punjab		-	90		100	100	55	-	97	-	90	-	86	66	66	-	-	3	0
Tewari et al.[a] (2018)[24]	New Delhi	ICU, Out-patients	93	67	-	-	-	54	60	80	-	83	67	-	47	47	20	-	0	-
Banerjee et al. (2018)[28]	Uttar Pradesh	ICU	92	83	-	-	91–92	-	90	89	77	85	83	86	70	85	0	-	0	-
Kaur et al. (2018)[27]	Punjab	ICU	74.2	87	-	-	-	97	95	89	-	92	90	92	60	68	-	-	3	4
Kumari et al.[a] (2019)[25]	New Delhi	ICU, wards	-	93	-	-	95	-	-	96	80	80	92	92	90	90	8	-	2	-
John et al. (2020)[31]	Tamil Nadu	ICU	-	97	-	-	-	97	-	-	96	92	96	96	99	99	-	-	1	-
Khamari et al. (2020)[37]	Andhra Pradesh	Wards	-	71	-	-	78	57	78	64	57	-	-	-	64	-	0	14	14	-
Kalal et al. (2020)[15]	Karnataka	ICU	-	85	85	-	95	85	-	91	-	87	85	87	95	95	-	-	-	-

[a]Retrospective studies. The numbers are expressed in % and are rounded to the nearest 10. Amox + clav, amoxicillin + clavulanic acid; Pip + taz, p iperacillin + tazobactum; Cefo + sulb, cefoperazone + sulbactum; 1st generation includes cefazolin, cephalexin; 2nd gen includes cefoxitin, cefuroxime; 3rd gen includes ceftazidime, ceftriaxone, cefoperazone, cefotaxime, cefpodoxime (extended spectrum cephalosporins); 4th gen includes cefepime (extended spectrum cephalosporins). FQs, fluoroquinolones, Cipro, ciprofloxacin; Levo, levofloxacin; TMP-SMX, trimethoprim–sulphamethoxazole; AGs, aminoglycosides; Amika, amikacin; Genta, gentamicin; Tige, tigecycline; Tet, tetracycline; Mino, minocycline; Poly-B, polymixin B; ICUs, intensive care units; NICU, neonatal intensive care unit; PICU, pediatric intensive care unit.

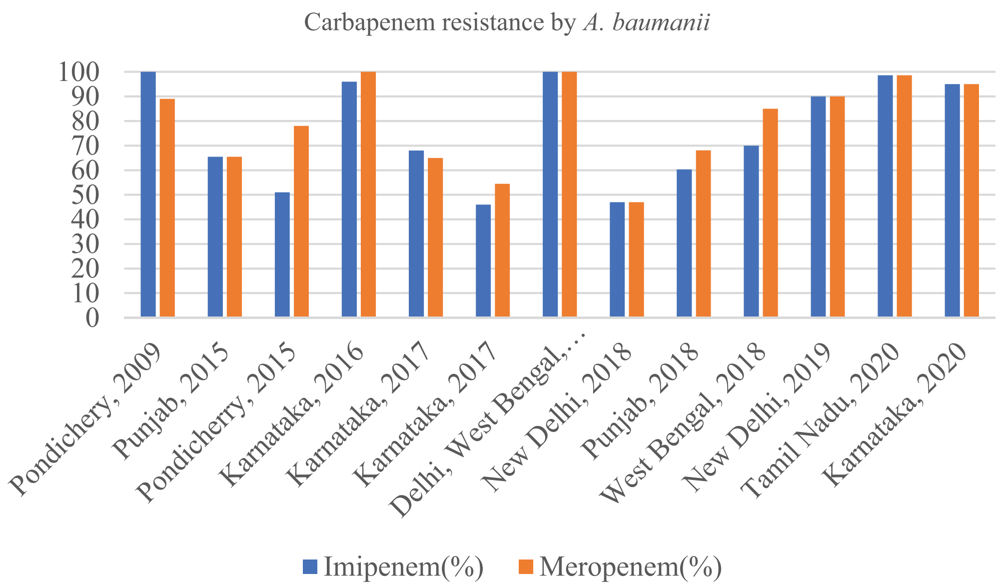

Figure 2. Carbapenem resistance by *A. baumanii* across India at different time periods.

2.2. Molecular characterization of AMR-related gene in MDR A. baumannii in the various Indian health care settings

Being a natural transformant, *Acinetobacter* genus is bestowed with the genetic arrangement for the rapid development of AMR. The resistance against carbapenems is, in itself, sufficient to define an *A. baumannii* as highly resistant [48]. *A. baumannii* has genes encoding a variety of carbapenem-hydrolyzing beta-lactamases under the Amber classification, namely (a) class A-ESBL (extended-spectrum β-lactamases) type (*blaTEM, blaPER, GES-11, KPC-2, KPC-3,KPC-4 and KPC-10*), (b)class B-MBL (Metallo-β-lactamases) type (*blaIMP-1, blaVIM-2, blaSIM-1* & *blaNDM*) and (c) class D-OXA type (*blaOXA*-23-like, *blaOXA*-24-like or *blaOXA*-40- like, *blaOXA-51-like, blaOXA-58-like, blaOXA-143-like* & *blaOXA-235-like*) [49]. The class D -OXA-type carbapenemases (also called oxacillinase type) are the most prevalent carbapenemases in MDR *A. baumannii*. Although *blaOXA-23* and *blaOXA-51* genes are probably most prevalent among *CRAB* isolates reported globally, India has documented more than one type of β-lactamases [50]. Although most frequently reported in Enterobacteriaceae, *blaNDM* genes have also been detected in MDR *A. baumannii*. The first case of *blaNDM* gene in *Acinetobacter* spp. was reported in 2010 from India and now this gene containing species is found in nearly 40 countries. In Latin America, *blaNDM gene* was first described in 2011 in *Klebsiella pneumoniae* from Guatemala and Colombia. *blaNDM* was also detected in other South American countries in 2012. NDM-1 was first found in *CRAB* isolate in Brazil in 2013. Most cases are directly linked to India, Pakistan, Bangladesh, or the Balkans region [51]. *Acinetobacter* spp. isolated from bedrails, door handles, surfaces of monitors, and washbasins in hospitals harbored *blaNDM*-1 gene highly resistant to imipenem [28].

Interestingly, several 'resistance Islands' have recently been identified among MDR isolates of *A. baumannii*, that either belongs to International clone 1 or 2 (IC1 or IC2). MDR *Acinetobacter* spp. have been globally disseminated in association with the successful clonal lineage *A. baumannii* (IC2). Both *CRAB* and carbapenem-susceptible *A. baumannii* IC2 are found in Asian countries [52].In India, *CRAB* IC2 is more common than IC1, IC,7, IC8. One such Indian study in 2019 showed the prevalence of *CRAB* IC2, IC1, IC7, IC8 as 44%, 8%, 1, 13% respectively [23]. India has very few reports on phenotypic and molecular characterization of *A. baumannii* isolates. However, studies suggest that *blaOXA* coding genes (class D-OXA-type carbapenamases) are most prevalent, followed by *blaIMP, blaVIM*, and *blaNDM*, under class B-MBL type and *blaTEM* under class A-ESBL types [12,15,28,37,40,53] (Table 3)

Colistin resistance is increasing alarmingly among clinical isolates of *A. baumannii*. Vijaykumar et al. using whole-genome sequencing observed that overexpression of ISAba1-driven eptA was responsible for colistin resistance among Indian isolates of colistin-resistant *A. baumannii* [18].

2.3. Prevalence of different β-lactamases expressed by AMR-related genes in MDR A. baumannii across India

Production of β-lactamases constitutes the major reason for resistance to β-lactams. For instance, ESBLs, MBLs, and most commonly, oxacillinases are the major contributors. The prevalence of ESBLs, AmpC β-lactamase, and MBL-expressing isolates vary across different geographic regions. Multiple studies from India between 2008 and 2020 reported the prevalence of AmpC and ESBLs β-lactamase-producing isolates in 20% to 99% and 4% to 83%, respectively [28,29,40,54,55]. AmpC β-lactamase enzymes provide resistance to many β-lactam antimicrobials, except carbapenem [12]. Reports indicate that 7.5% to 96.6% *A. baumannii* isolates express MBL [27,40,56]. Multiple β-lactamases expressing *A. baumannii* isolates have also been reported across India. Co-production of ESBLs + AmpCs and ESBLs + AmpCs + MBLs in MDR *A. baumannii* have also been reported in two studies ranging from 20%-29% and 0–7%, respectively (Table 4)[40,54].

Table 3. Depicts the distribution of different antibiotic coding genes found in *A. baumannii* that expresses β-lactmases from various studies across India.

		Class A (ESBLs type)		Class B and Class D (Carbapenamases type)				
				blaNDM-1			blaOXA	
Author and year	Location	blaTEM	blaPER-1		blaVIM-1	blaIMP-1	blaOXA-23	blaOXA-51
Khamari et al. 2020[37]	Andhra Pradesh	64	100	43	-	-	14	-
Kalal et al. 2020[15]	Karnataka	64	-	13	64	16	98	96
Kumar et al. 2019[23]	Punjab, Haryana	-	-	29	-	-	98	4
Vijaykumar et al. 2019 [12]	Tamil Nadu	7	57	17	-	-	100	100
Banerjee et al. 2018[28]	Uttar Pradesh	-	-	34 and 27[a]	51	89	93	-
Pragasam et al. 2016[56]	New Delhi, Punjab, Tamil Nadu, Pondicherry	16	54	22	3	-	98	100
Rynga et al. 2015[40]	Delhi	-	-	-	50	6		-
Saranathan et al. 2015[59]	Pondicherry	-	14	-		41	57	100
Rajarajacholan et al. 2009[43]	Pondicherry	-	-	-	-	42	-	-
Rao et al. 2008[53]	Pondicherry	-	20	-	-	-	-	-

The numbers are expressed in % and are rounded to the nearest 10. [a]blaNDM identified from environmental samples.

Table 4. β-Lactamases produced by clinical isolates of *A. baumannii* shown in different studies across India.

Author and year	Location	ESBLs	AmpCs	MBLs	Co-production of ESBLs +AmpCs	Co-production of ESBLs+AmpCs +MBLs
Banerjee et al. 2018[28]	West Bengal	79	-	-	-	-
Kaur et al. 2018[29]	Punjab	28	-	45	-	-
Pragasam et al. 2016[56]	New Delhi, Punjab, Tamil Nadu, Pondicherry	71	-	25	-	-
Ryngya et al. 2015[40]	Delhi	38	99	29	-	7
Saranathan et al. 2015[59]	Pondicherry			55%		
Singla et al. 2014[54]	Haryana	4	60	-	29	-
Gupta et al. 2015[41]	Pune	32	-	14	-	-
Rajarajacholan et al. 2009[43]	Pondicherry	-	-	71	-	-

The numbers are expressed in % and are rounded to the nearest 10. ESBLs, extended spectrum β-lactamases, AmpCs, ampicillinase C; MBLs, metallo-β-lactamases.

2.4. A report on biofilm producers in MDR A. baumannii strains

A. baumannii also produces biofilms encoded by the *bfmR* gene. Biofilm-producing organisms are far more resistant to antibiotics than organisms that do not. In some extreme cases, the concentrations of antibiotics required to achieve bactericidal activity against biofilm-producing isolates can be three- to fourfold higher, depending on the spp. and drug combination. *A. baumannii* ranks second only to *P. aeruginosa* among the biofilm-producing, nosocomial, non-fermentative aerobic, Gram-negative bacteria. Biofilm production in *A. baumannii* promotes increased colonization and persistence leading to more frequent DA-HAI [57]. Studies have shown that MDR *A. baumannii* strains that readily form biofilms are mainly associated with DA-HAI, whereas non-biofilm producing strains are associated with respiratory infection or colonization [20].

There is a positive correlation between drug resistance and biofilm production. Investigations by Badave et al. (2015) and Khamari et al. (2020) have revealed that 62% and 71.4% of MDR *A. baumannii* isolates were strong biofilm producers [37,58]. In another study, Gurung et al. showed that 50% of *A. baumannii* isolates produced biofilms with 73% among them highly resistant to AGs, third-GCs and imipenem, showing an MDR pattern [57]. Saranathan et al. in 2015 reported that 62% and 32% of *A. baumannii* strains were moderate and robust biofilm producers, respectively [59]. Rao et al. in 2008 reported 62% of isolates were biofilm producers and all were 100% resistant to imipenem [53]. *A. baumannii's* virulence potential to form biofilms helps escalate AMR and acquire genetic material from unrelated genera, making it a versatile and challenging adversary to control and eradicate [1].

3. Device-associated-hospital-acquired infections (DA-HAI) due to MDR *A. baumannii*

DA-HAI induced bacteremia and sepsis are the leading cause of prolonged hospital stay, barrier nursing and patient isolation, morbidity, mortality, treatment-cost escalation, and reduction in bed availability. HAI–induced mortality associated with VAP, CLABSI, and CAUTI varies from 15% to 50%, 12% to 25%, and 7% to 35%, respectively. Khan et al. in 2017 reported the range of VAP, CLABSI, and CAUTI rates in India has been 6–32, 3.4–16, and 2.1–11.3 per 1000 device-days, respectively [36]. *A. baumannii* is one of the most common Gram-negative pathogens for device-related infections such as VAP, CLABSI and CAUTI. Kumar et al. have shown that 18.3% *CRAB* caused DA-HAI in which 40% of *A. baumannii* strains caused VAP [60]. Different studies across India have shown a higher rate of VAP than any other DA-HAI caused by *A. baumannii* (Table 5). In developing countries, including India, *Acinetobacter* has emerged as the 4th leading cause of VAP and a significant cause of other DA-HAI's. The known risk factors for *Acinetobacter* colonization/infection are prolonged hospital or ICU stay, longer duration of MV and catheterization, previous admission to another unit, immunosuppression, debilitation, previous use of third-GCs, and blood transfusions [10,33]. Infection surveillance analysis is an imperative prerequisite for quality care and prevention of DA-HAI. Several studies have shown that routine surveillance can reduce the incidence of *A baumannii* by as much as 30% and reduce

Table 5. Device-related infection by *A. baumannii*.

Author and year	Location	VAP	CLABSI	CAUTI
John et al. 2020(26)	Tamil Nadu	74	60	-
Vekataraman et al. 2018(42)	Tamil Nadu, Maharashtra, New Delhi, Haryana, Telangana	32	12	0
Kumar et al. 2018(60)	New Delhi	40	-	-
Khan et al. 2017(36)	Delhi, West Bengal, Maharastra	67	28	5
Gupta et al. 2016(33)	Karnataka	33	-	-
Saranathan et al 2015(59)	Pondicherry	33	11	16
Datta et al. 2014(35)	Punjab	41	27	10
Mathai et al. 2012(10)	Punjab	59	4	-

The numbers are expressed in % and are rounded to the nearest 10. VAP, ventilator-associated pneumoniae; CLABSI, central line-associated blood stream infection; CAUTI, catheter-associated urinary tract infection.

morbidity, mortality, and healthcare costs proportionately. However, due to the lack of formal surveillance in developing countries like India, the rate of HAI is high and compliance with hand hygiene is low [35].

4. Relationship between coronavirus disease 2019 (COVID-19) with *A. baumannii*

Since the WHO declared the COVID-19 pandemic on 12 March 2020, many patients infected with COVID-19 required admission to the ICU. Bacterial co-infection or super-infections are common in viral respiratory infections and are important causes of morbidity and mortality. Within the first few days of COVID-19 infection, critically ill patients often develop pulmonary dysbiosis or respiratory tract distortion, which can progress further to a secondary bacterial or fungal infection. *A. baumannii* as an opportunistic infection is chiefly associated with nosocomial infections, particularly in ICUs, where the incidence has increased over time. Data from several countries have shown that co-infection with *A. baumannii* secondary to COVID-19 infections ranges from 1% to 90%. Unfortunately, there are no such data from India. COVID-19 pandemic has promoted overuse of antimicrobials, especially broad-spectrum antibiotics, in critically ill patients, aggravating the AMR threat [16,61].

5. Mortality

MDR *A. baumannii*, has become a significant hospital pathogen, with attributable mortality ranging from 8% to 35%, depending on the strain [3]. Patients infected with MDR *A. baumannii* were 2 to 3 times more likely to die than patients with non-MDR infections [22]. Bacteremia and VAP caused by XDR *A. baumannii* were associated with >50% mortality rates [15]. There is considerable debate on the relationship between clinical *A. baumannii* infection and mortality. Different studies have different patient outcomes, undermining consensus [1]. Results are likely to be confounded by the difficulty in determining whether death resulted from *A. baumannii* infection or potential co-infections. However, recent studies have concluded that MDR *A. baumannii* infections alone are responsible for increased mortality. John et al. in 2020 reported the overall mortality rate of 50.3% in which 57.6% death were from VAP and CLABSI caused more by MDR than non-MDR *A. baumannii* infections [31]. Another report from Gandra et al. in 2019 showed overall mortality of 13.1%. Mortality rate was higher in patients infected with *A. baumannii* (29.0%) and lower in patients infected with *E. coli* (8.8%) and *Staphylococcus aureus* (11.0%) [22]. Ghanshani et al. in 2015 also reported an overall mortality rate of 13.9% due to HAI, with the highest death [11.8%] observed in *A. baumannii* infection [62]. Mathai et al. and Saranathan et al. (in 2012 and 2015, respectively) reported mortality rate of 21% and 70.21% among patients with *A. baumannii* infections, in addition to inappropriate antibiotic treatment and poor ventilator outcomes after the detection of infection [10,59]. Mortality from *Acinetobacter* infections varies from 30% to 55%, especially after VAP in *Acinetobacter* infections. Tiwari et al. and Sudhaharan et al. have recorded 28.7% and 22% mortality in 2013 and 2015, respectively [39,63]. Another study in 2016 reported a 34.4% mortality from VAP caused by *A. baumannii*, in which 41.2% of deaths were attributed to colistin-resistant *A. baumannii* isolates (53.6%) [33]. However, it is essential to recognize that these patients were in a critical condition with comorbidities, potentially raising the mortality rate. The severity of the underlying disease, pneumonia, MDR strains, septic shock, disseminated intravascular coagulation, MV, and inappropriate antimicrobial treatment has been implicated in raising the mortality from *Acinetobacter* infections [64].

6. Cost

Estimates of disease burden is important because they help compare the relative impact of a wide range of conditions using common metrics. In addition, decision-makers can use disease burden estimates in allocating scarce healthcare resources toward treatment or prevention. Healthcare cost and mortality are intertwined. For example, healthcare costs could potentially be lower in patients who die of HAI because premature death terminates all healthcare services. On the other hand, many studies have documented the increase in healthcare costs associated with end-of-life treatment, suggesting that mortality can be associated with higher overall cost. Mortality–cost relationships make sense only when cost and mortality are reported together [64]. *Acinetobacter* spp. alone is responsible for approximately 10.6% of mortality due to MDR infections, with an estimated cost ranging from USD 33,510 to USD 129,917 [25]. There is a shortage of studies related to cost associated with hospital-acquired *A. baumannii* infections in India. A six-month study by Asim et al. in 2016 reported an increase in both median length of stay (20 days) and treatment cost of MDR *A. baumannii* infections (USD 1874) than non-*Acinetobacter* infections (12 days and USD 1016) [65]. A retrospective case-control study reported that the total cost incurred due to HAI was four to five times higher than those without HAI. Most of the cost associated with MDR *A. baumannii* infection (30.6%)

occurred from imipenem use and extended hospital stay [63].

Medication costs increase with AMR. The rise of MDR strains in *Acinetobacter* infections will also necessitate last-resort antimicrobials such as colistin, raising medication costs further. More than 80% of the Indian population pay out-of-pocket because of poor health insurance coverage. Medical bills can even push families into bankruptcy [65]. The cost burden of *Acinetobacter* infections has a significant socioeconomic impact on victims' families. Effective screening programs for *Acinetobacter* can offset the burden of cost due to the *Acinetobacter* infections.

7. MDR *A. baumannii* infection control

In 2006 and 2007, the US Healthcare Infection Control Practices Advisory Committee (HICPAC) and the CDC published guidelines for the management of MDR pathogens and isolation precautions in healthcare settings [66]. Despite these guidelines, in 2011, the European Antimicrobial Resistance Surveillance System Network (EARS-Net), which includes 29 European countries, reported a pan-European increase of antimicrobial resistance in Gram-negative pathogens under surveillance [67]. Despite guidelines on specific measures for detecting and controlling transmission, *A. baumannii* nosocomial outbreaks continue, along with the emergence of MDR, and XDR strains of *A. baumannii*, further raising the stakes [21].

There are several reports of effective control and eradication of MDR *A. baumannii* using a combination of techniques. However, current management of MDR *A. baumannii* prevention, infection and outbreaks is based on observational studies and pharmacodynamic modeling. While many guidelines contain broad areas of agreement, there are inconsistencies between guidelines, reflecting the limited evidence in published literature. Management is based around source control, including antimicrobial stewardship program (AMSP), hand hygiene, contact precautions, education, and effective environmental cleaning. Many studies suggest mandatory closure of hospital units to control outbreaks. This can come at a great cost. Studies from the USA and London has shown the estimated cost of *A. baumannii* outbreak to be USD 350,000. The most significant cost component is associated with lost bed days through extended patient stays or ward closures, reducing surgical ward capacity [21].

A major reason for the emergence of *Acinetobacter* spp. is environmental. There are very few alternatives to tackle MDR *Acinetobacter* infections, other than strict adherence to available strategies [28].

In India, Sharma et al. in 2014 reported that implementing infection control measures in a burn unit, reduced the incidence of MDR *A. baumannii* from 1 to 2 cases every 2 to 3 days to 1 to 2 cases per 14 days. Remedial measures such as (a) restricted antibiotic usage based on antimicrobial susceptibility report, (b) empirical therapy for short periods, (c) high-level disinfectants (hydrogen peroxide with silver nitrate), and (d) proper disposal of soiled linen and other biomedical waste from wards, diminished the presence of MDR *A baumannii* in environmental samples from the dressing room and transport trolley [68].

Strict implementation of contact precautions, personnel protective equipment, hand hygiene, environmental disinfection, isolation of patients, education and training of both healthcare professionals and patients' attendants are necessary for effective control of MDR *A. baumannii* outbreaks.

8. Conclusion

The high burden of MDR isolates and *CRAB* harboring multiple carbapenemase-encoding genes, especially primarily associated with VAP, is a major concern in Indian healthcare settings. Endemicity of the *A. baumannii* in the ICU environment, intense antimicrobial selection pressure, and lack of AMSP are the probable causes. Reducing the risk of infection in hospitalized patients requires DA-HAI surveillance because it effectively describes and addresses the importance and characteristics of the threatening situation created by HAIs. Most importantly, superinfection or co-infections secondary to other infections such as COVID-19 must be studied and controlled. Surveillance must be followed by the implementation of practices aimed at DA-HAI prevention and control. The intervention module includes educating staff, creating a catheter insertion chart, mandating daily reports on catheterization, emptying of urine bag in addition to maintaining standard practices such as keeping the urobags always below the bladder and implementing a checklist to ensure the adherence to evidence-based guidelines for preventing CLABSI, CAUTI, and VAP. Nurses should be empowered to stop catheter insertion procedures that violate guidelines. The occurrence of MDR or *CRAB* has enhanced the virulence potential of the isolates, compromising treatment and raising mortality rates. *CRAB* infections compels clinicians to opt for a combination of last-resort antibiotics such as polymyxins and tigecycline in many parts of India. The increasing trend toward 'Watch group' antibiotic usage necessitates hospital AMSP implementation to facilitate a shift to 'Access group' antibiotic usage. Extensive research on the geographical pattern of resistance, molecular characterization, cost, and mortality due to *A. baumannii* will facilitate the allocation of healthcare resources toward implementing antibiotic usage guidelines, treatment, and prevention. The dearth of research in producing novel antimicrobials and India's poor surveillance system have enfeebled control of MDR *A. baumannii* infections and its consequences such as escalated cost, hospital stay and mortality. The Indian Council of Medical Research (ICMR) initiated Antimicrobial Resistance Surveillance & Research Network (AMRSN) has not been functionally established across the Indian states. Eradicating or containing MDR *A. baumannii* infections requires the enforcement of specific policy-directed interventional methods.

9. Expert opinion

Over recent years, *Acinetobacter baumanni* complex (A b complex), *Acinetobacter baumannii* (A. baumannii) in particular, has become a 'red-alert' human pathogen, largely due to its unique ability to develop resistance to all currently available antibiotics and also persist in a harsh environment, provoking problematic outbreaks and life-threatening infections. Most common risk factors of MDR *A. baumannii* infection are previous antibiotic exposure, use of medical devices such as mechanical ventilation and catheters, extended ICU/hospital stay, and severity of illness. A. baumannii is an important nosocomial organism associated with relatively high morbidity and mortality. Community-acquired infections (including pneumonia and bacteremia) are less common.

World Health Organization (WHO) classified MDR strains and carbapenem-resistant *A. baumannii* (CRAB) as 'Priority one' culprit on the global list of AMR micro-organisms requiring the need to develop novel antibiotics. MDR and extensive-drug resistant (XDR) *A. baumannii* have been reported worldwide with limited treatment options. While carbapenems are the last-resort therapy for *A. baumannii*, more than 90% *A. baumannii* are resistant to carbapenems. Device-associated-hospital-acquired infection (DA-HAI)- induced sepsis is the leading cause of extended hospital stay, necessitating barrier nursing, patient isolation, last-resort antibiotics, increased morbidity, and mortality, thereby increasing treatment cost and diminishing bed availability. Globally, mortality rates exceeding 50% have been reported due to CRAB infections like bloodstream infections (BSI) and ventilator-associated pneumonia (VAP).

Research funding and support for understanding AMR mechanisms, validation of current infection control practices, and new antibiotic molecule development must be prioritized. The knowledge of genetic structure of CRAB is invaluable to illustrate the epidemiology, surveillance and understand the global distribution. Molecular characterization using whole-genome sequencing techniques help identify genes (blaNDM, blaPER, etc.) and international clones (IC) responsible for emergence of AMR.

The most common antibiotics for treating CRAB include combinations employing tigecycline, colistin, and sulbactam. Unfortunately, increased usage of colistin for treating Acinetobacter infection results in AMR to this last-line drug. There is hardly any antibiotic that treats CRAB infections. Current intensive efforts address all the AMR mechanisms of *A. baumannii*, with the objective of identifying the most promising therapeutic scheme. Bacteriophage-based therapeutic approaches have been labeled as effective but require additional clinical evidence.

Many reports from the USA show that routine surveillance reduces the incidence of *A. baumannii* up to 30%. Suboptimal surveillance in developing countries like India, poor compliance with hand hygiene, mandatory closure of wards, unrestricted prescription of last-resort antibiotics, etc. have made MDR and CRAB, major hospital pathogens, with mortality ranging from 8% to 35%, depending on the strain. Few of the Indian states such as Kerala (2018), Madhya Pradesh (2019), and New Delhi (2020) have submitted the state action plan to the central government, but initiating and implementing AMSP among Indian hospitals is still nascent.

Most recommendations for public health, agricultural and veterinary fields mandate limiting restrictive use of antibiotics to slow down a natural evolution toward AMR and subsequent spread in the downstream environment in a 'One Health' setting. A critical study of *A. baumannii*, suggest that adopting a combination of multiple strategies such as rational use of antibiotics, following hygienic practices, prevention of device-associated-health care infections (DA-HAI), understanding the genetic components associated with the AMR, establishing AMSP in health care facilities, practicing barrier nursing, producing, and administering novel antibiotics, would reduce AMR or limit its epidemic spread. To successfully synchronize these strategies would be a challenge to researchers and physicians.

In conclusion, India faces many unresolved challenges in developing and implementing antibiotic policies to control MDR *A. baumannii*. Topmost priorities for the Government and hospital authorities include enforcing stringent infection control measures, surveillance, and antimicrobial stewardship to prevent further spread and emergence of more virulent and resistant strains. Most importantly, government should increase research funds for investigating the mechanism of antibiotic resistance and designing novel antibiotics that evade resistance.

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Abbreviations

WHO = World Health Organization
CDC = Centers for Disease Control and Prevention
A. b complex = *Acinetobacter baumannii* complex
CRAB = Carbapenem-resistant *Acinetobacter baumannii*
MDR = Multidrug-resistant
AMR = Antimicrobial resistance
USA = United States of America
DA-HAI = Device-associated-hospital-acquired infection
VAP = Ventilator-associated pneumonia
CLABSI = Central line-associated bloodstream infection
CAUTI = Catheter-associated urinary tract infection
AMRSN = Antimicrobial Resistance Surveillance & Research Network

ICMR = Indian Council of Medical Research
INICC = International Nosocomial Infection Control Consortium
HAI = Hospital-acquired infections
ICU = Intensive care units
BSI = Blood stream infections
UTI = Urinary tract infections
AGs = Aminoglycosides
FQs = Fluoroquinolones
GCs = Generation cephalosporins
OmpA = Outer membrane protein-A
MBL = Metallo-β-lactamase
NDM = New Delhi metallo-β-lactamase
IC2 = International clone 2
OXA = Oxacillinase
XDR = Extensive drug-resistant
MOSER = Multicenter Observational Study to Evaluate Epidemiology and Resistance
COVID-19 = Coronavirus disease 2019
ESBLs = Extended spectrum beta-lactamases
KPC = *Klebsiella pneumoniae* carbapenamases
blaVIM = blaVerona imipenemase metallo-β-lactamase
blaIMP = blaImipenemase IMP metallo-β-lactamase
blaTEM = blaTemoniera metallo-β-lactamase
blaSHV = blasulphydryl variable metallo-β-lactamase
blaOXA = blaOxacillin-Hydrolyzing β-lactamase
bfmR gene = biofilmR gene
USD = United States dollar
AMSP = Antimicrobial stewardship program
HICPAC = Healthcare Infection Control Practices Advisory Committee
EARS-Net = European Antimicrobial Resistance Surveillance System Network

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